

Conflating Antecedents and Formative Indicators: A Comment on Aguirre-Urreta and Marakas

Edward E. Rigdon

Department of Marketing, J. Mack Robinson College of Business, Georgia State University, Atlanta, Georgia 30303,
erigdon@gsu.edu

Jan-Michael Becker

Department of Marketing and Brand Management, University of Cologne, Cologne 50923, Germany,
j.becker@wiso.uni-koeln.de

Arun Rai

Center for Process Innovation and Department of Computer Information Systems, J. Mack Robinson College of Business,
Georgia State University, Atlanta, Georgia 30303, arunrai@gsu.edu

Christian M. Ringle

Institute of Human Resource Management and Organizations, Hamburg University of Technology (TUHH),
Hamburg 21073, Germany; and University of Newcastle, Newcastle, NSW 2308, Australia, ringle@tuhh.de

Adamantios Diamantopoulos

Chair of International Marketing, University of Vienna, 1090 Vienna, Austria, adamantios.diamantopoulos@univie.ac.at

Elena Karahanna

Department of Management Information Systems, Terry College of Business, University of Georgia,
Athens, Georgia 30602, ekarah@uga.edu

Detmar W. Straub

Professor, Georgia State University, Atlanta, Georgia 30303; Visiting Professor, Fox School of Business,
Temple University, Philadelphia, Pennsylvania 19122; and Visiting Professor, Korea University Business School,
Korea University, Seoul 136-701, Korea, dstraub@gsu.edu

Theo K. Dijkstra

University of Groningen, 9712 CP Groningen, The Netherlands, t.k.dijkstra@rug.nl

Aguirre-Urreta and Marakas [Aguirre-Urreta MI, Marakas GM (2014) Research note—Partial least squares and models with formatively specified endogenous constructs: A cautionary note. *Inform. Systems Res.* 25(4):761–778] aim to evaluate the performance of partial least squares (PLS) path modeling when estimating models with formative endogenous constructs, but their ability to reach valid conclusions is compromised by three major flaws in their research design. First, their population data generation model does not represent “formative measurement” as researchers generally understand that term. Second, their design involves a PLS path model that is misspecified with respect to their population model. Third, although their aim is to estimate a composite-based PLS path model, their design uses simulation data generated via a factor analytic procedure. In consequence of these flaws, Aguirre-Urreta and Marakas’ (2014) study does not support valid inference about the behavior of PLS path modeling with respect to endogenous formatively measured constructs.

Keywords: formative indicators; partial least squares; endogenous constructs

History: Vijay Mookerjee, Senior Editor; Sunil Mithas, Associate Editor. This paper was received on December 18, 2013. Published online in *Articles in Advance* September 15, 2014.

1. Purpose of the Rejoinder

In their research note, “Partial Least Squares and Models with Formatively Specified Endogenous Constructs: A Cautionary Note,” Aguirre-Urreta and Marakas (2014; hereafter A&M) seek to evaluate

whether partial least squares (PLS) path modeling is a valid statistical method for estimating models with formative endogenous constructs. They conclude that, “we show here PLS is not an adequate approach to modeling scenarios where a latent variable of interest

is endogenous to other latent variables in the research model in addition to its own observed formative indicators" (A&M, p. 776). The objective of this rejoinder is to show that the analysis and conclusions of A&M are invalid because the research design described in their paper as the basis for their conclusions has three major flaws. First and most fundamentally, their population data generation model does not represent "formative measurement" as researchers generally understand that term; instead, A&M conflate the roles of "antecedent" and "indicator" for a formative construct. Second, A&M estimate a PLS path model that is misspecified with respect to their population model; thus, A&M induce discrepancy, which confounds their ability to reach conclusions about the performance of PLS path modeling. Third, A&M generate simulation data using a factor analytic procedure even though PLS path modeling is a method of composites: this introduces known biases in parameter estimates that cannot be overcome without special procedures. As a consequence of these flaws, A&M are not in a position to reach valid conclusions about the behavior of PLS path modeling with respect to constructs composed by formative indicators and endogenous to other constructs.

2. The Arguments Against A&M's Approach

2.1. First Flaw: Population Data Generation Model Inconsistent with "Formative Measurement"

A&M's statistical population model does not represent "formative measurement" as researchers generally understand the term (e.g., Bagozzi 2011, Bollen and Davis 2009, Diamantopoulos and Winklhofer 2001, Diamantopoulos et al. 2008, Diamantopoulos 2011, Petter et al. 2007). What are formative indicators? Although literature on the topic has a long history, broad interest in formative indicators stems largely from a small number of papers, each of which has been cited thousands of times. According to Fornell and Bookstein (1982, p. 442), "when constructs are conceived as explanatory combinations of indicators (such as "population change" or "marketing mix") that are determined by a combination of variables, their indicators should be formative." Thus, formative indicators are observed variables that together form or compose a construct.

In PLS path modeling, every conceptual variable is represented by a composite of observed variables (Fornell and Bookstein 1982, Rigdon 2012). Thus, when A&M specify a PLS path model where three observed variables are indicators of a composite, their PLS path model *correctly* represents a formative indicator relationship. Using A&M's symbols, this relationship can be represented as

$$\eta_1 = w_1x_1 + w_2x_2 + w_3x_3, \quad (C1)$$

where $w_1 - w_3$ are indicator weights. In addition, A&M include in their model a predictor construct, labeled ξ_1 . The equation for this relationship (again using A&M's symbols) can be specified as

$$\eta_1 = \gamma_1\xi_1 + \zeta_1, \quad (C2)$$

where ζ_1 is a structural error term.

Consider these two equations. The first represents η_1 as a weighted sum of three observed variables, $x_1 - x_3$. The second indicates that ξ_1 predicts this *weighted composite*. In other words, ξ_1 predicts the weighted composite of variance formed by the three observed variables.

The basis of A&M's research design, however, is a very different model. Their equation (AM1), which they acknowledge is the basis for all of their analysis and the source of their simulation data, is as follows (A&M, p. 763):

$$\eta_1 = \gamma_1x_1 + \gamma_2x_2 + \gamma_3x_3 + \gamma_4\xi_1 + \zeta_1. \quad (AM1)$$

In A&M's (p. 765) words, "Equation (1) states that a formatively specified latent variable is a function of its manifest indicators, a latent predictor, and a disturbance term." Here, the three observed variables and the predictor construct *jointly* predict η_1 . Thus, it cannot be said that x_1 , x_2 , and x_3 "form" η_1 unless one also says that ξ_1 and the error term ζ_1 also "form" η_1 . One might as well say that *any* dependent variable is formed by its predictors. This is certainly not what researchers generally understand as "formative measurement." In A&M's Equation (AM1), the three observed variables are simply predictors, like ξ_1 . They are not formative indicators of η_1 .

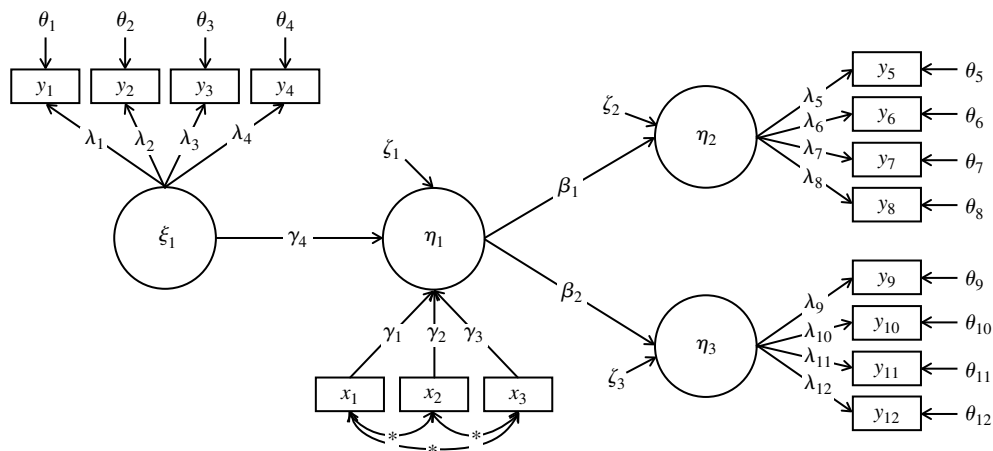
Thus, we conclude that A&M's research objective of evaluating endogenous formative constructs and their research model are not compatible.

2.2. Second Flaw: PLS Path Model Misspecified with Respect to Population Model

Having specified a population model that is incompatible with formative measurement, A&M then specify a PLS path model that is inconsistent with their population model. A&M primarily present their PLS path model in graphical form, and a reader might perceive a good match between A&M's population model (in their Figure 1, p. 763; reproduced here as Figure 1) and their PLS path model (in their Figure 2, p. 764; reproduced here as Figure 2).

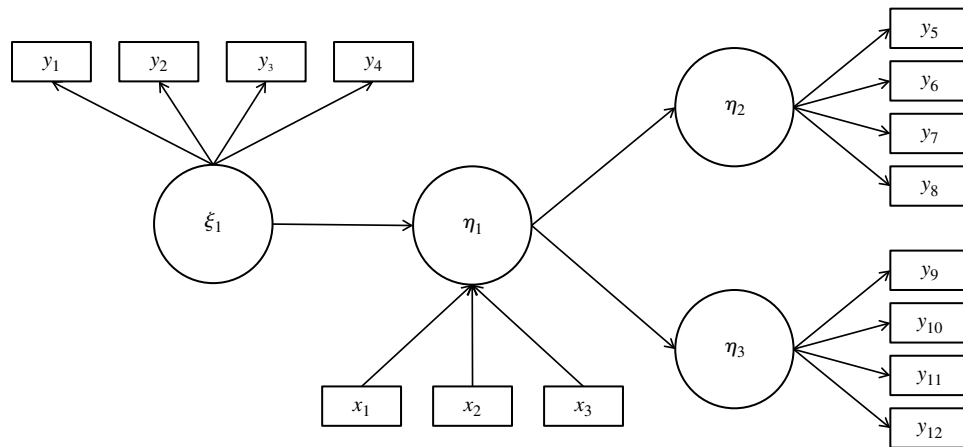
However, the similarity between the two diagrams cannot be taken to mean that there is a match between the two statistical models, due to a difference in graphical conventions. Within PLS path modeling, conceptual variables are represented statistically by composites. The only permitted direct paths between one of these composites and any observed variables

Figure 1 A&M's Population Model Diagram



Source. Reproduced from Aguirre-Urreta and Marakas (2014, Figure 1, p. 763).

Figure 2 A&M's PLS Path Model Diagram



Source. Reproduced from Aguirre-Urreta and Marakas (2014, Figure 2, p. 764).

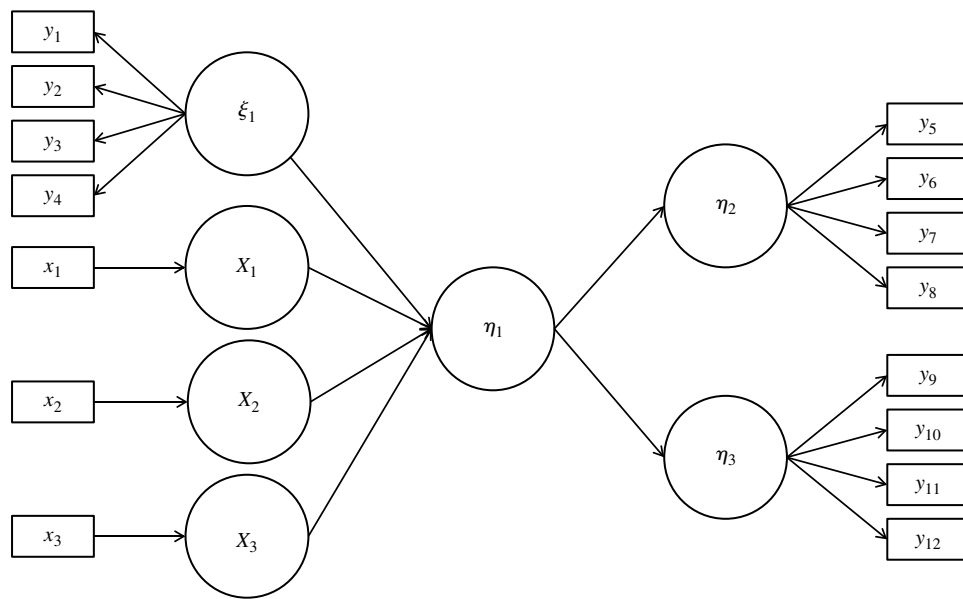
are the paths between a composite and its component observed variables. Thus, in the diagram of A&M's *PLS path model* (Figure 2), the links between their η_1 composite and the observed variables x_1 , x_2 , and x_3 indicate that x_1 , x_2 , and x_3 are the three observed variable components of η_1 . But in A&M's *diagram of their population model* (Figure 1), the very same graphical representation indicates that x_1 , x_2 , and x_3 are predictors of η_1 and not components. So, even though the two diagrams appear similar, A&M's Figure 1 provides a plausible match to their population model, but A&M's Figure 2 does not. This means that A&M's PLS path model does not match the population model that is the basis for their analysis, and this mismatch prevents A&M from reaching valid conclusions about the performance of PLS path modeling.

A better match to A&M's population model would be a PLS path model as depicted in Figure 3. In this PLS path model, there are four structural predictors for η_1 , just as A&M's population model specifies.

Each of the observed variables $x_1 - x_3$ is the sole component of a composite that serves as a structural predictor for η_1 . Of course, from a conventional PLS path modeling perspective, there is a serious problem with this model. Whereas factor-based structural equation modeling (SEM) uses common factors to represent conceptual variables, PLS path modeling uses composites. Composites need components, and η_1 does not have any. Thus, the model shown in Figure 3 cannot be estimated using conventional PLS path modeling.¹ But this should not obscure the point that A&M draw conclusions about the behavior of PLS path modeling from a PLS path model that does not match their population model.

¹ There are ways to model η_1 as a hierarchical latent variable (second-order type construct) of either ξ_1 and $x_1 - x_3$ or η_2 and η_3 (e.g., Becker et al. 2012, Ringle et al. 2012, Wetzels et al. 2009), but this is not, we believe, what A&M intended.

Figure 3 PLS Path Model That Reflects A&M's Population Model



2.3. Third Flaw: Simulation Data Generated Using a Factor Analytic Procedure Even Though PLS Path Modeling Is a Method of Composites

Factor models and composite models are generally not equivalent. Although composite methods can often approximate results from factor models (Velicer and Jackson 1990) and vice versa (Steiger 1990), using composite methods to replicate factor models leads to known estimation biases (McDonald 1996, Wold 1982), which cannot be overcome without special procedures (Dijkstra and Schermelleh-Engel 2014). A&M (p. 766) make clear that they used a factor-based SEM procedure to generate their simulation data. Given that they planned to estimate their model using a composite method, they should have used a composite method (such as that described by Becker et al. 2013) to generate their data.

There is a further distinction between factor-based methods and composite-based PLS when it comes to conceptual variables that are both formatively measured and structurally endogenous. In factor-based SEM, for every dependent variable, there is a single equation that collects all of the dependency relationships for that one variable. So, in factor-based SEM, even though there may be important conceptual distinctions between predictor factors and formative indicators, all of the dependency relations for a given dependent variable are collected and represented in the same single equation, just as specified in A&M's Equation (AM1).

By contrast, in PLS path modeling there is a multistage estimation process. All conceptual variables in PLS path modeling are represented by weighted composites. Relations between composites and their indicators are distinct from relations between composites

and other composites. In PLS path modeling, the two equations (C1) and (C2) remain two *separate* equations. Unlike factor-based SEM, and unlike A&M's Equation (AM1), there is no forced merger of these conceptually different relations into a single equation.

In using factor-based SEM software (i.e., Mplus 3.0) to generate the data for the simulation analysis, A&M may have been misled by exactly this distinction between methods. Specifying a model in factor-based SEM as it appears in A&M's Figure 1 will lead to a statistical model that resembles their population model equations; as previously noted, however, this is not consistent with "formative measurement" as it is generally understood in the literature. Perhaps this error would not have occurred if A&M had created their data using a composite-based approach.

3. Conclusion

A&M used a misspecified equation as the fundamental basis for their analysis and for the generation of their simulation data. In addition, they used a misspecified PLS path model as the basis for drawing conclusions about the behavior of PLS path modeling. Finally, they used a factor-based procedure instead of a composite-based procedure to generate simulation data to be used in estimating composite-based PLS path models. Given the flaws described here, there is no sound evidence in A&M's paper to support their conclusion that PLS path modeling does not perform well in modeling endogenous formative constructs. In fact, their approach does not enable them to draw *any* valid conclusions about PLS path modeling with endogenous formatively measured constructs.

References

- Aguirre-Urreta MI, Marakas GM (2014) Research note—Partial least squares and models with formatively specified endogenous constructs: A cautionary note. *Inform. Systems Res.* 25(4): 761–778.
- Bagozzi RP (2011) Measurement and meaning in information systems and organizational research: Methodological and philosophical foundations. *MIS Quart.* 35(2):261–292.
- Becker JM, Klein K, Wetzels M (2012) Hierarchical latent variable models in PLS-SEM: Guidelines for using reflective-formative type models. *Long Range Planning* 45(5–6):359–394.
- Becker JM, Rai A, Rigdon EE (2013) Predictive validity and formative measurement in structural equation modeling: Embracing practical relevance. *Proc. 34th Internat. Conf. Inform. Systems, Milan*, <http://aisel.aisnet.org/icis2013/proceedings/ResearchMethods/5/>.
- Bollen KA, Davis WR (2009) Causal indicator models: Identification, estimation, and testing. *Structural Equation Modeling: A Multidisciplinary J.* 16(3):498–522.
- Diamantopoulos A (2011) Incorporating formative measures into covariance-based structural equation models. *MIS Quart.* 35(2): 335–358.
- Diamantopoulos A, Winklhofer HM (2001) Index construction with formative indicators: An alternative to scale development. *J. Marketing Res.* 38(2):269–277.
- Diamantopoulos A, Riefler P, Roth KP (2008) Advancing formative measurement models. *J. Bus. Res.* 61(12):1203–1218.
- Dijkstra TK, Schermelleh-Engel K (2014) Consistent partial least squares for nonlinear structural equation models. *Psychometrika* 79(4):585–604.
- Fornell C, Bookstein FL (1982) Two structural equation models: LISREL and PLS applied to exit-voice theory. *J. Marketing Res.* 19(4):440–452.
- McDonald RP (1996) Path analysis with composite variables. *Multivariate Behavioral Res.* 31(2):239–270.
- Petter S, Straub D, Rai A (2007) Specifying formative constructs in information systems research. *MIS Quart.* 31(4):623–656.
- Rigdon EE (2012) Rethinking partial least squares path modeling: In praise of simple methods. *Long Range Planning* 45(5–6): 341–358.
- Ringle CM, Sarstedt M, Straub D (2012) A critical look at the use of PLS-SEM in *MIS Quarterly*. *MIS Quart.* 36(1):iii–viii.
- Steiger JH (1990) Some additional thoughts on components, factors and factor indeterminacy. *Multivariate Behavioral Res.* 25(1): 41–45.
- Velicer WF, Jackson DN (1990) Component analysis versus common factor analysis: Some issues in selecting an appropriate procedure. *Multivariate Behavioral Res.* 25(1):1–28.
- Wetzels M, Odekerken-Schröder G, van Oppen C (2009) Using PLS path modeling for assessing hierarchical construct models: Guidelines and empirical illustration. *MIS Quart.* 33(1): 177–195.
- Wold HO (1982) Soft modeling: The basic design and some extensions. Jöreskog KG, Wold H, eds. *Systems Under Indirect Observation: Causality, Structure, Prediction (Part II)* (North-Holland, Amsterdam), 1–54.

Copyright 2014, by INFORMS, all rights reserved. Copyright of Information Systems Research is the property of INFORMS: Institute for Operations Research and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.